PERFORMING RIGS

AutoPilot Firmware Validation Report

Document	AP-VVR-001
Revision	Rev A
Product	AutoPilot
Hardware Revision	Hardware Revision 1 (bench unit)
Firmware Revision	Autopilot_firm29-06.rar · build group build-2026-06-29 (received 2026-06-29)
Validation Date	July 9, 2026
Prepared By	Nathan Destro, Performing Rigs
Location	Bench / shop

Validation Summary

Validation Status	**OPEN**
Functional Result	**CONDITIONAL PASS**

Validation Status — Initial bench validation of the June 2026 trigger-focused beta. Validation remains open pending off-site / field-representative trigger testing and resolution (or documented disposition) of startup and INFO-screen regressions.

Functional Result — Several intended improvements were observed on the bench (200 mA minimum current, trigger-type selector, INFO voltage readouts). Field-style false-trigger failures were **not** reproduced in the shop. Release is **not** approved: Quick Configuration is missing at startup (AP-006), and INFO diagnostics cannot be used during an actual move (AP-007 / AP-008).

1. Executive Summary

Noxon sent a **trigger-focused beta** (Autopilot_firm29-06 / build-2026-06-29) intended to reduce false triggering and improve diagnostics. On the Performing Rigs shop bench (Hardware Revision 1), several improvements look real: lower minimum motor current, a new trigger-type selector, and voltage / signal-strength readouts on the INFO screen.

However, the startup **Quick Configuration** wizard no longer appears, and the new INFO diagnostics **cannot be used during an actual move** because the UI immediately switches to a "RUNNING — press button to stop" screen. Trigger-input errors reported from the field were **not reproduced** in the shop, so further testing away from the bench is required before calling this release ready.

2. Evidence Summary

Validation Item	Result
Bench flash of firm29-06 (915 MHz autopilot bin)	**PASS**
Minimum current reduced to 200 mA	**OBSERVED**
Trigger signal type selector present	**OBSERVED**
INFO screen trigger voltage / signal strength	**OBSERVED**
Quick Configuration at startup	**FAIL** (missing)
INFO readable during programmed move	**FAIL** (forced RUNNING screen)
Field false-trigger (AP-001) reproduction	**NOT OBSERVED** (shop)
Off-site / noisy-environment validation	**DEFERRED** (see VVR-002)

Test environment. Performing Rigs shop bench; AutoPilot Rev 1; firmware via Noxon update_firmware.bat firmware_autopilot915.bin. Trigger scenarios: partial / not yet matching field failure modes.

3. Findings

F-001 — Quick Configuration missing at startup AP-006

Motor quick-setup at power-on no longer appears. Startup **Quick Configuration** flow is removed or gated, breaking the operator workflow documented for initial channel setup.

Severity: High

Status: Open

F-002 — INFO screen exits on trigger during move AP-007

Cannot watch signal levels while testing a real move. When a programmed move runs, the UI **forces the RUNNING screen**; INFO diagnostics are unavailable during motion.

Severity: Medium

Status: Open

F-003 — INFO formatting (Signal Strength line break) AP-009

Signal lines are hard to read. Request a **line break after "Signal Strength"** before the trigger reading row.

Severity: Low

Status: Open

F-004 — Diagnostic mode requested AP-008

Allow INFO / trigger readout **without executing moves** — dedicated container or "info mode" that ignores trigger events for measurement.

Severity: Medium

Status: Open (enhancement)

F-005 — Field false-trigger not reproduced on bench AP-001

Unexpected repeated triggering / unintended movement with trigger cable connected remains a field report. Shop test this session was inconclusive / not reproduced.

Severity: Critical (field class)

Status: Open — not reproduced this session

Positive observations (not defects)

Item	Note
Minimum current	Reduced from **1.0 A to 200 mA** — lowers hold torque / heat during idle; verify stall / recovery under load
Trigger signal selector	New **input signal type** setting (relay vs 12 V vs 24 V per Noxon roadmap) — required for Rev 2 / mitigation strategy
INFO diagnostics	Exposes trigger voltage / signal strength — supports TP-006 validation protocol when usable

4. Questions for Noxon

Q-001 At the new 200 mA minimum, what is the **lowest voltage** at which outputs energize, and for **how long** after power-on?

Q-002 When do discussions begin on **jog**, runtime display, and other Category B roadmap items?

Q-003 What **objective criteria** decide which field units get Rev 2 boards vs firmware-only mitigation?

Q-004 What test matrix did Noxon run internally before shipping firm29-06?

5. Recommendations

R-001 Do not ship to customers on this build until AP-006 is resolved or explicitly deprecated with documentation; AP-007 / AP-008 are addressed or a field-diagnostics workaround is documented; and field-trigger tests complete (AP-001 reproduction or signed-off mitigation evidence).

R-002 Internal use: Continue bench + off-site validation; keep the Oct 2025 stable build as rollback reference (Autopilot_firm11-12.rar / build group build-2025-10-27).

6. Remaining Validation

From EMS TestingProtocols TP-001–TP-007 (status as of later sessions noted where complete):

• [x] Off-site / electrically noisy environment (TP-006) — **2026-07-22 VVR-002:** Relay Contact + 12/24 V (9 V) PASS after recovery; intermittent post-reset watch remains (AP-001)
• [x] Relay vs 12 V vs 24 V modes with known-good wiring — **partial PASS** (see VVR-002)
• [ ] Long trigger cable runs
• [x] Alternate trigger devices (dry contact, Dragonframe path, etc.) — MRMC dry contact + 9 V PASS (VVR-002)
• [ ] Recovery after disconnect/reconnect (TP-007)
• [ ] A/B vs Oct 2025 stable build (build-2025-10-27) on **same hardware**

Engineering Assessment

This package is a useful **trigger / diagnostics beta** for bench learning, not a customer ship candidate. Treat observed INFO and current-limit improvements as promising; treat missing Quick Configuration and INFO-during-move as release blockers until dispositioned. Carry AP-001 forward until field-representative evidence closes or mitigates it.

Source: Bench notes and screenshot Screenshot 2026-07-09 at 8.58.13 AM.png in Engineering Development.

Revision History

Rev	Date	Description
A	2026-07-09	Issued — initial bench validation of firm29-06 / build-2026-06-29 (formatted twin 2026-09-11)